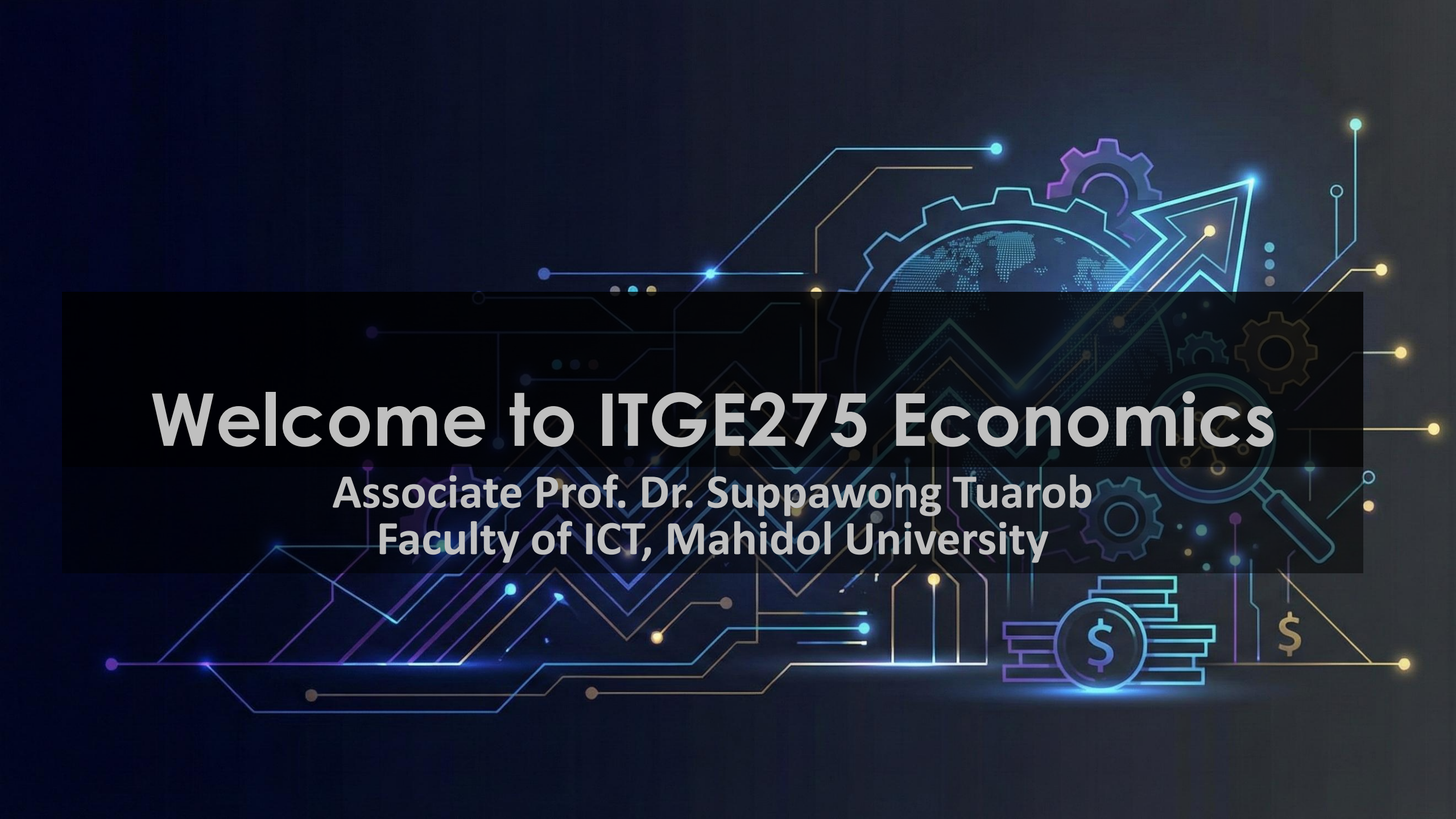


Welcome to ITGE275 Economics

Associate Prof. Dr. Suppawong Tuarob
Faculty of ICT, Mahidol University



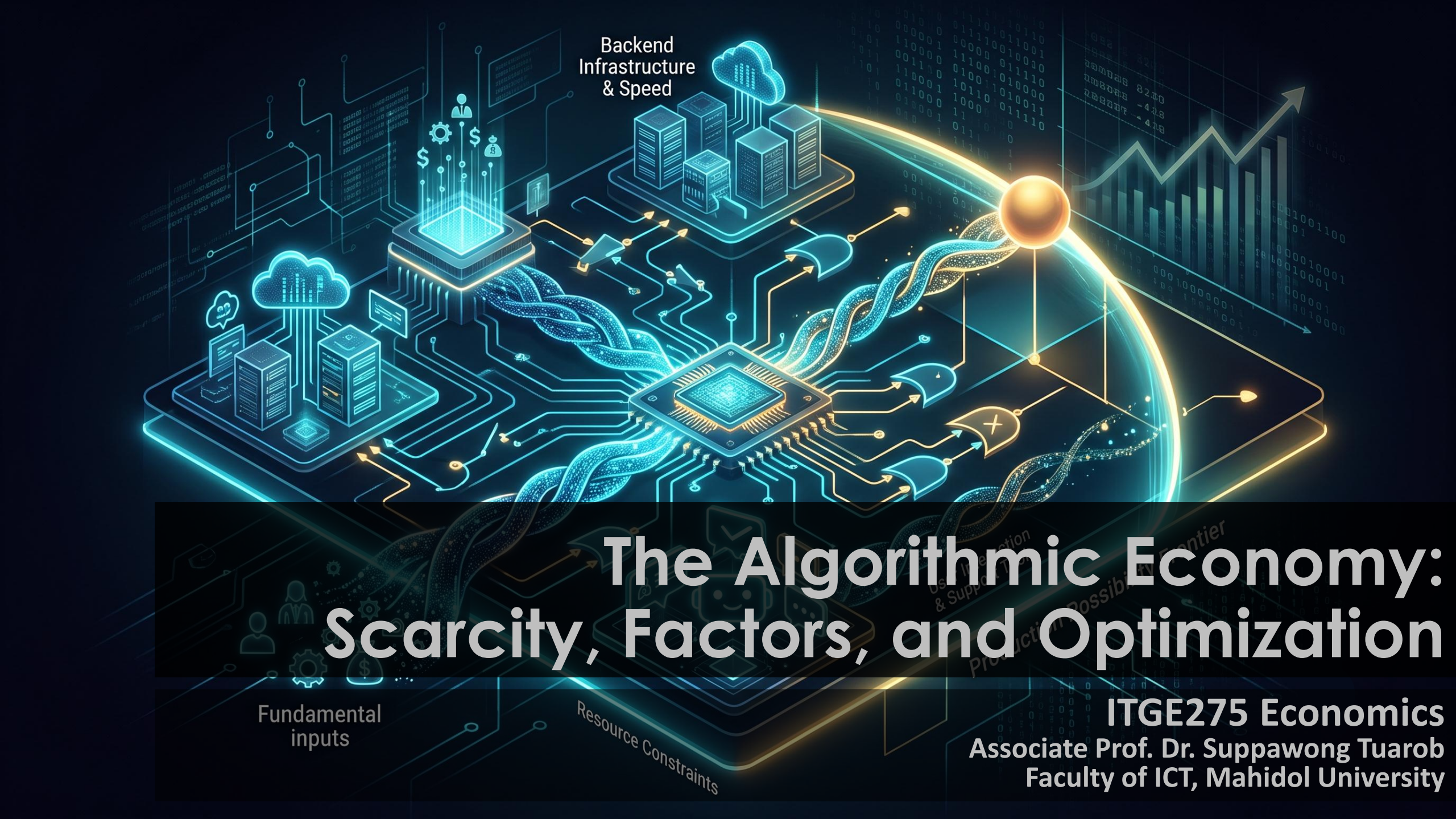
Curse Logistics

- Lecture: MON 8:30 (😢 , Me Too) - 10:30, IT302
- Instructor: Assoc. Prof. Dr. Suppawong Tuarob
  - *Email:* suppawong.tua@mahidol.ac.th
- LA: Mr. Prahyat Puangjaktha (prahyat.pua@student.mahidol.ac.th)
- Evaluation:
  - 40% Homework (Due in one week)
  - 30% Midterm
  - 30% Final
- Grade: OSU



Class Management

Time	Detail
8:30 – 8:40	Attendance
8:40 – 9:30	Lecture
9:30 – 10:00	Notebook Session
10:00 – 10:30	Interactive Session/QA



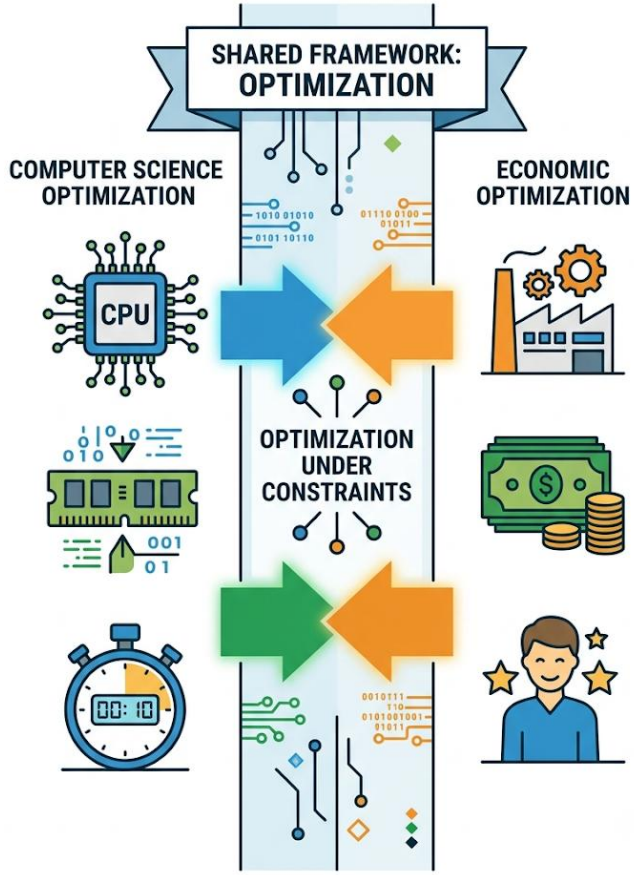
The Algorithmic Economy: Scarcity, Factors, and Optimization

ITGE275 Economics

Associate Prof. Dr. Suppawong Tuarob
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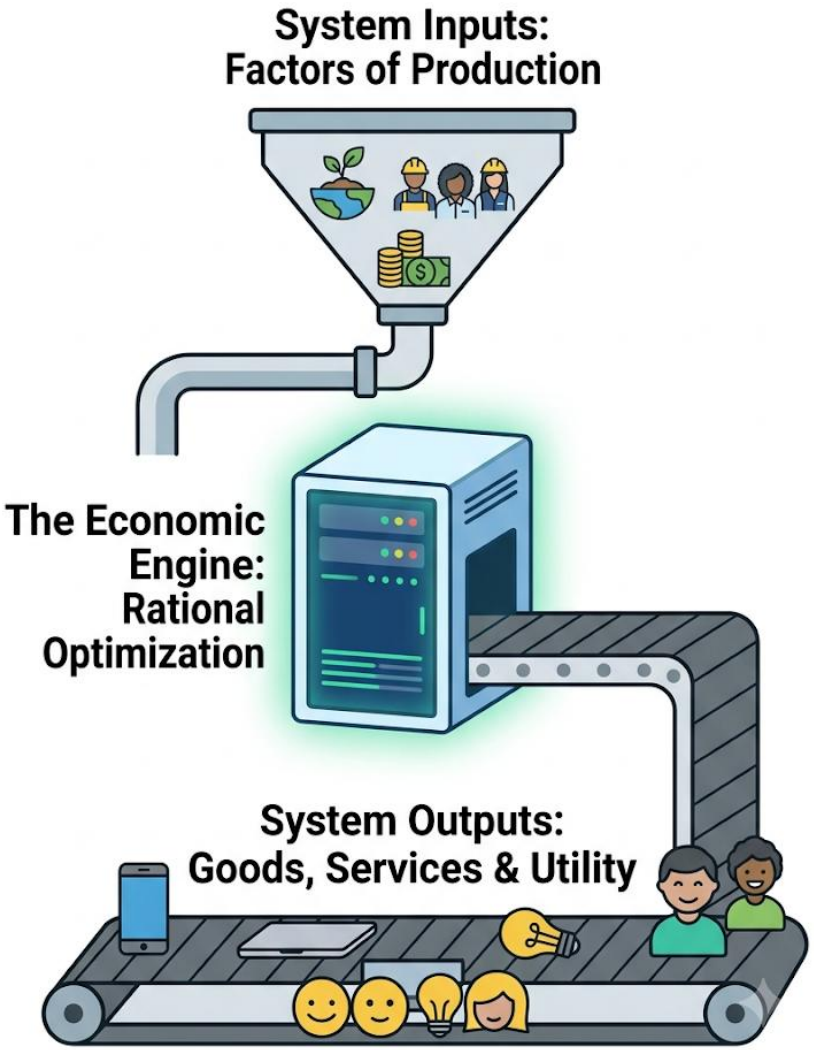
The Core Bridge: Algorithms and Economics

- **The Computer Science Approach:** When designing an algorithm, computer scientists seek to maximize computational speed and memory efficiency while operating under strict hardware limitations.
- **The Economic Approach:** Economics is the scientific study of allocating limited resources to maximize utility (human satisfaction) or corporate profit.
- **The Shared Framework:** Both disciplines are fundamentally defined as the science of optimization subject to unavoidable constraints.
- **Why This Matters to You:** Viewing an economy as a computational system allows you to model human behavior, business decisions, and resource distribution using familiar analytical logic.



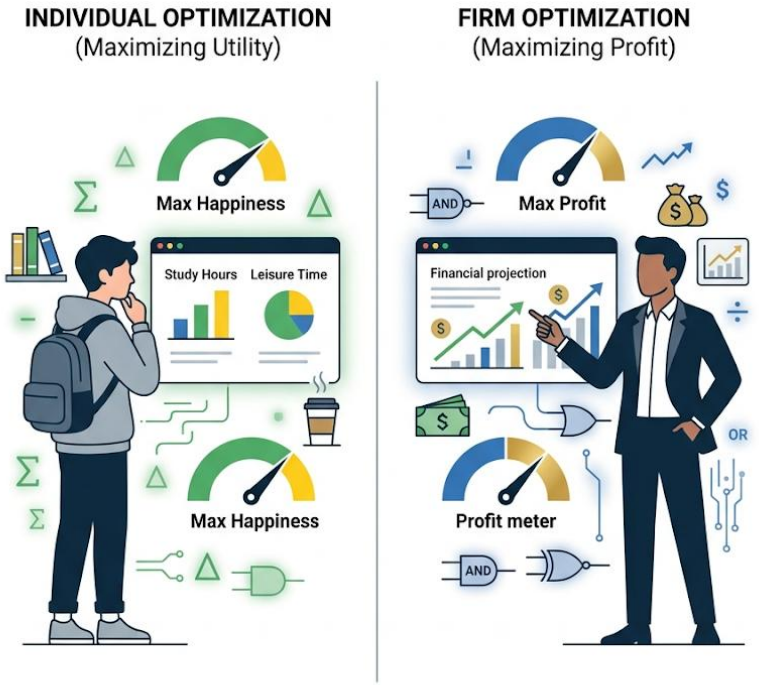
The Algorithmic Mindset

- **System Inputs:** In programming, algorithms require data inputs to function; in economics, productive systems require basic inputs known as Factors of Production.
- **System Rules:** Programs operate according to strict logical conditional rules; economic actors operate based on foundational behavioral assumptions.
- **System Outputs:** A successful script outputs solved data or completed tasks; a successful economy outputs valuable goods, services, and societal utility.
- **The Analytical Lens:** Throughout this course, we will analyze businesses and markets not as abstract theories, but as dynamic, quantifiable systems.



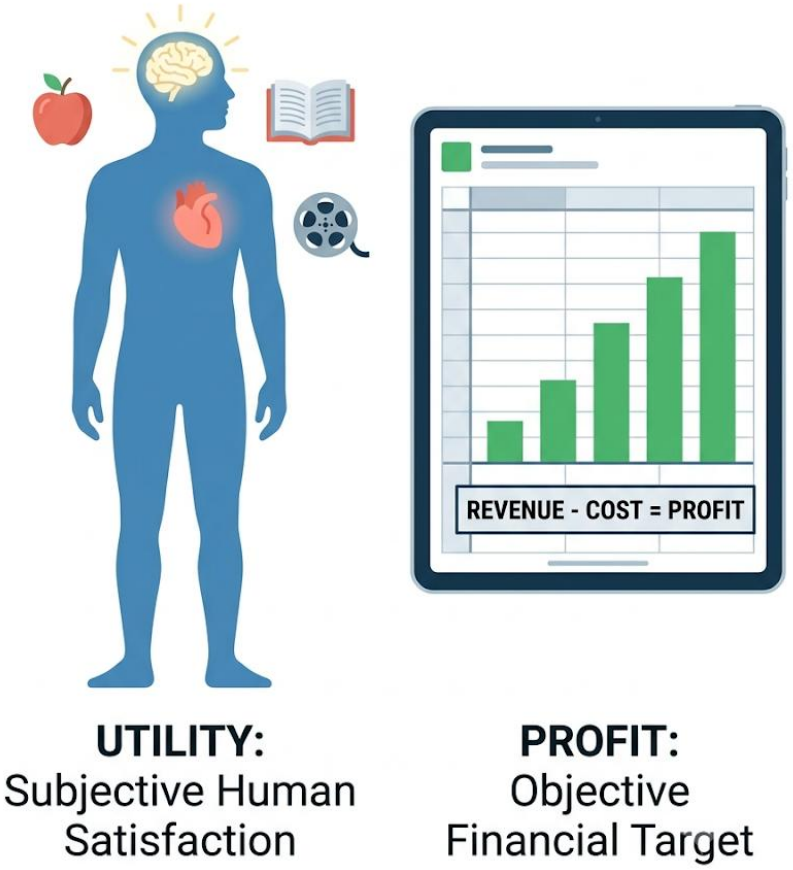
Foundational Economic Assumptions

- **The Rational Actor Model:** Economists build theoretical models by assuming that all participating agents make consistent, logical decisions.
- **Individual Optimization:** We assume that individual consumers always act rationally to maximize their personal utility and overall happiness.
- **Firm Optimization:** We assume that commercial businesses and firms always act rationally to maximize their net financial profits.
- **The Role of Constraints:** Neither individuals nor firms can achieve infinite satisfaction or profit; every decision must be made within the boundaries of available resources.



Deconstructing Utility and Profit

- **Defining Utility:** Utility is an economic term representing the subjective measure of satisfaction, value, or usefulness a person gains from an action or good.
- **Defining Profit:** Profit is an objective financial metric calculated by subtracting total production costs from total revenue generated.
- **The Objective Function:** In computational terms, utility and profit serve as the target objective functions that our economic algorithms attempt to maximize.
- **Predictable Behavior:** Because agents actively seek to maximize these metrics, their responses to changing incentives and resource limitations become mathematically predictable.



UTILITY:
Subjective Human Satisfaction

PROFIT:
Objective Financial Target

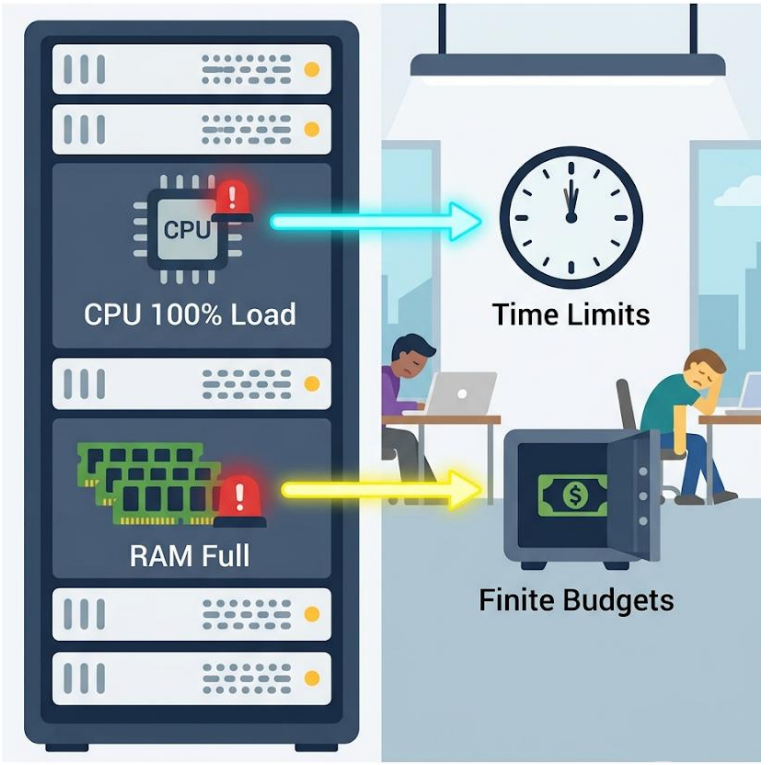
Redefining Scarcity

- **Beyond Sold-Out Goods:** Scarcity in economics is not merely a temporary retail shortage where a store runs out of a specific physical product.
- **The Universal Condition:** Scarcity is the fundamental, permanent condition where human wants and desires exceed the finite resources available to fulfill them.
- **The Catalyst for Choice:** Because resources are inherently scarce, systems and individuals are continuously forced to make choices about what to prioritize.
- **The Core Problem:** Without scarcity, there would be no trade-offs, no prices, and no need for the science of economics or optimization.



The CS Analogy: Hardware Scarcity

- **Time as Clock Cycles:** In daily life, time is limited to 24 hours a day; in computing, processor execution is limited by fixed CPU clock speeds.
- **Money as Memory:** A business has a finite operational budget; a software application has a strict allocation of RAM and disk storage.
- **Processing Power as Labor:** A startup has a limited number of human developers; a server farm has a limited number of computational nodes.
- **Shared Reality:** Just as an operating system must ruthlessly manage thread priority to prevent system crashes, an economy must allocate scarce inputs to prevent waste.



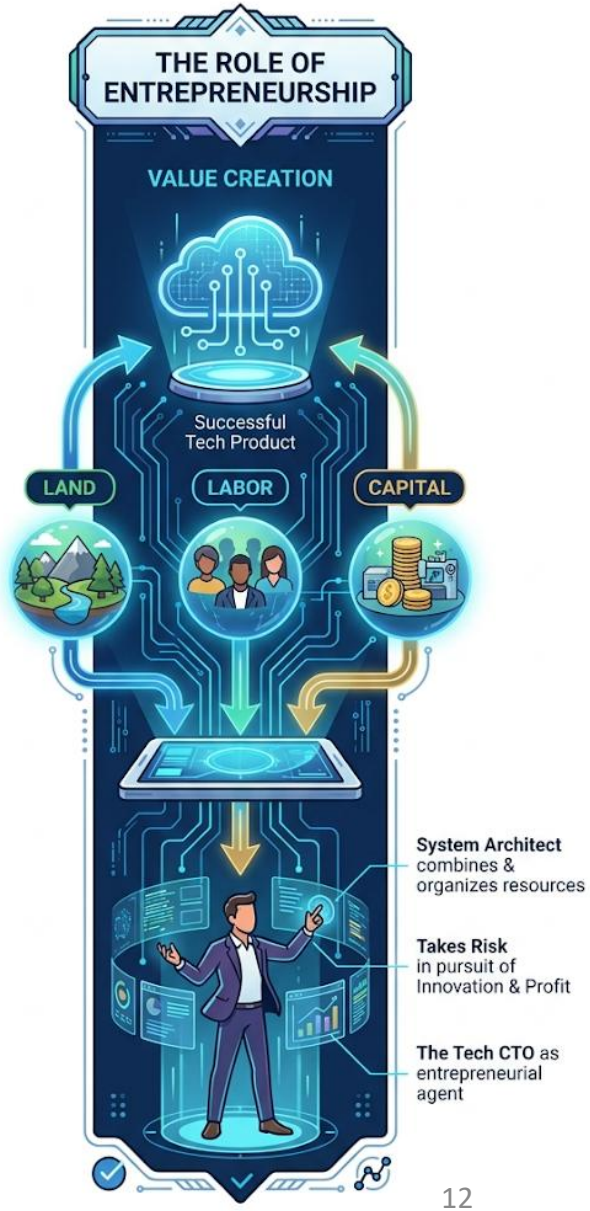
The Four Factors of Production

- **System Inputs:** Every good or service produced in an economy requires the combination of foundational inputs known as the Factors of Production.
- **Land (Natural Resources):** All physical physical space and naturally occurring resources used in production, such as real estate, water, minerals, and energy.
- **Labor (Human Effort):** The physical and mental effort contributed by human workers, including developers, engineers, designers, and support staff.
- **Capital (Man-Made Tools):** The machinery, computers, software licenses, buildings, and financial funds used to build other goods.



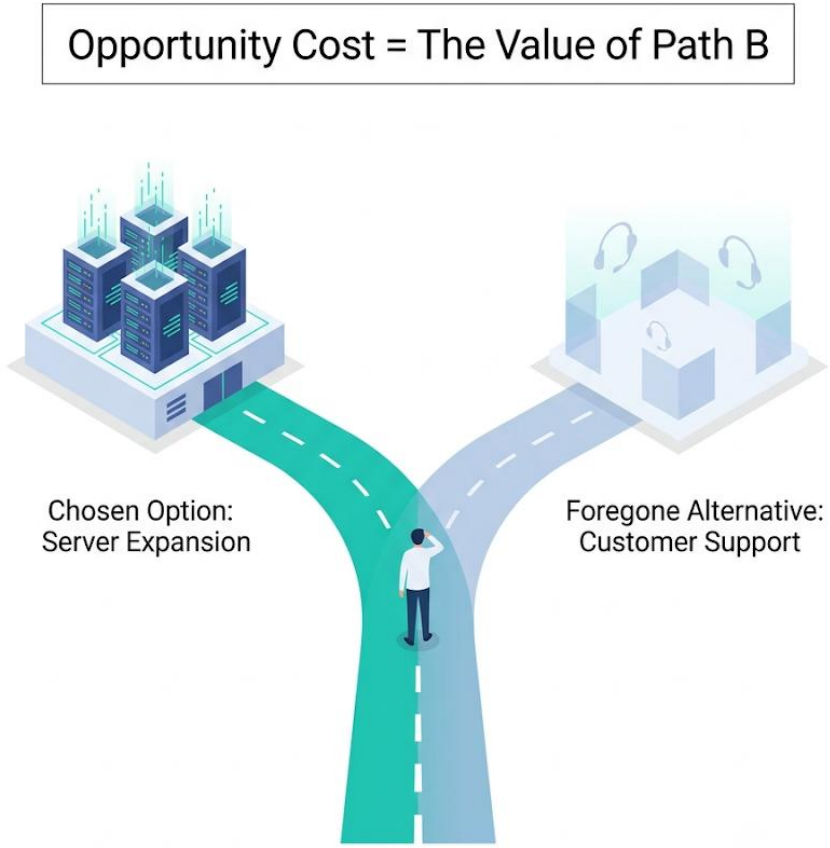
The Role of Entrepreneurship

- **The Fourth Factor:** While Land, Labor, and Capital represent raw resources, they remain completely inert without the fourth factor: Entrepreneurship.
- **The System Architect:** The entrepreneur is the visionary agent who combines, organizes, and manages the other three factors to create a productive firm.
- **Risk and Innovation:** Entrepreneurship involves taking assuming financial and operational risks in pursuit of innovation and future profit.
- **The Tech CTO:** In our algorithmic economy models, the Chief Technology Officer (CTO) or startup founder acts as the primary entrepreneurial optimizing agent.



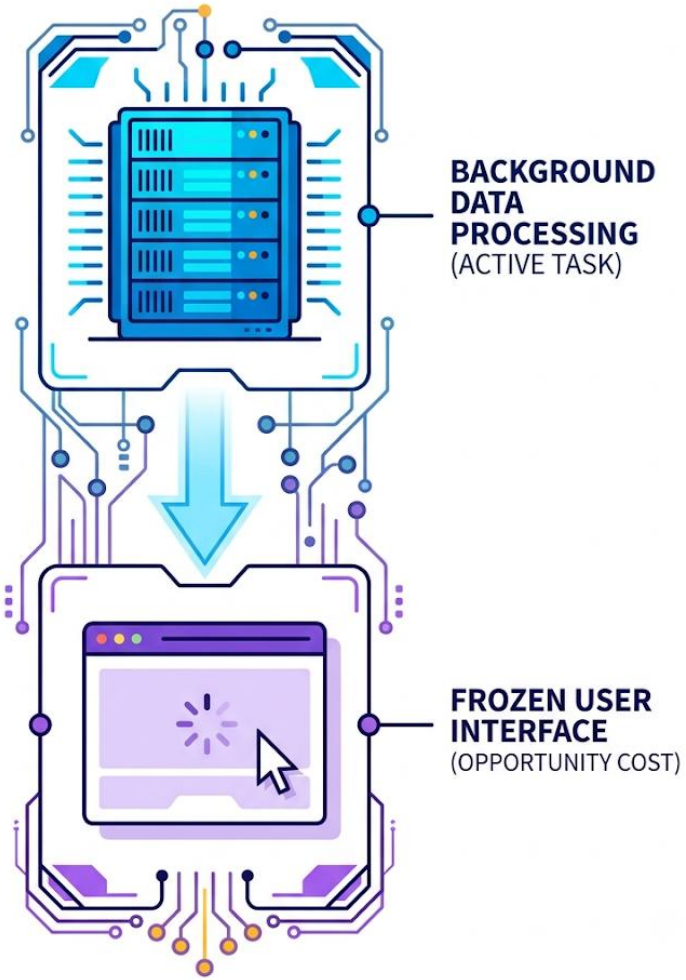
Opportunity Cost Defined

- **The True Cost of Choice:** In economics, the cost of acquiring something is not merely the money paid, but the intrinsic value of what you must give up.
- **Formal Definition:** Opportunity Cost is precisely defined as the value of the next best alternative foregone when a resource allocation decision is made.
- **The Invisible Expense:** When resources are fully utilized, every choice carries a hidden opportunity cost that must be accounted for in rational optimization.
- **No Free Lunch:** Because resources are scarce, choosing to allocate inputs toward Project A mathematically requires sacrificing the potential benefits of Project B.



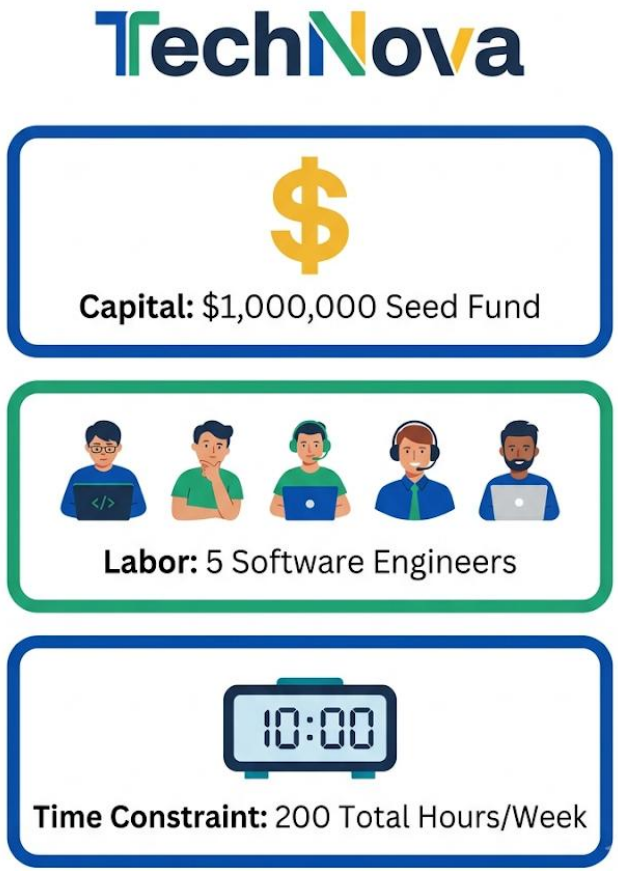
Opportunity Cost in Computer Science

- **CPU Core Allocation:** If a single-core CPU is actively executing a background rendering task, its opportunity cost is the immediate inability to process user interface inputs.
- **Memory Management:** When RAM is heavily allocated to storing large static datasets, the opportunity cost is the reduced memory available for dynamic application execution.
- **Bandwidth Saturation:** Utilizing network bandwidth for massive background system downloads carries the opportunity cost of increased latency for real-time user traffic.
- **Financial Capital:** A startup keeping $1,000,000 locked in idle inventory incurs the opportunity cost of the interest or growth that capital could have earned in a bank.



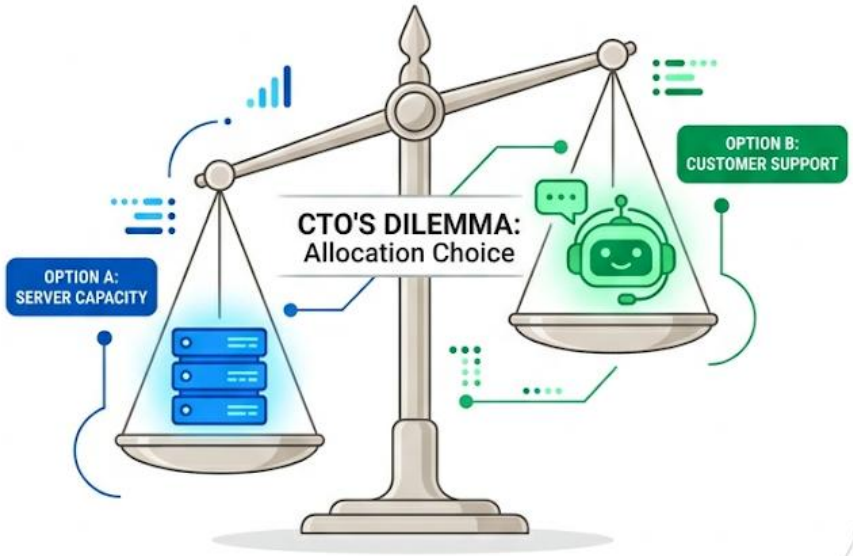
Case Study: "The Startup's Dilemma"

- **The Scenario:** You have just been appointed as the Chief Technology Officer (CTO) of a promising new software startup named "TechNova".
- **Your Fixed Capital:** The company has secured an initial seed funding round of exactly $1,000,000 to fund operations.
- **Your Fixed Labor:** You have hired a dedicated, highly skilled technical team consisting of exactly 5 software developers.
- **Your Fixed Time:** Each developer works a standard 40-hour week, giving your system a strict total resource constraint of 200 developer hours per week.



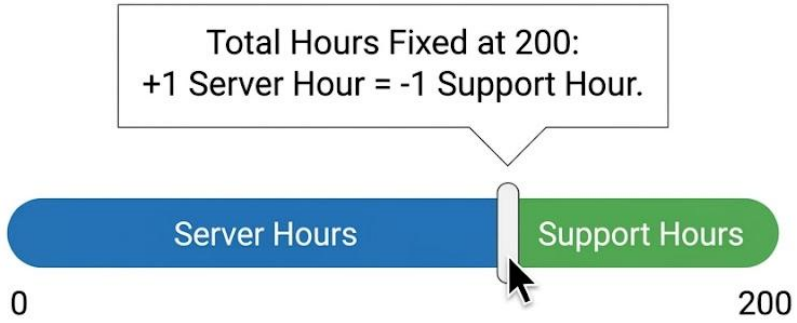
Competing Infrastructure Priorities

- **Option A - Server Capacity:** Deploying developer hours to build robust backend architecture, database indexing, and scalable server infrastructure.
- **Option B - Customer Support:** Deploying developer hours to build user-facing tools, automated ticketing systems, and AI-driven support chatbots.
- **The Strategic Conflict:** Server capacity ensures high system stability and performance during traffic spikes, while customer support drives user satisfaction and retention.
- **The CTO's Dilemma:** You cannot simultaneously build both systems to maximum capacity; you must decide how to divide your 200 developer hours between them.



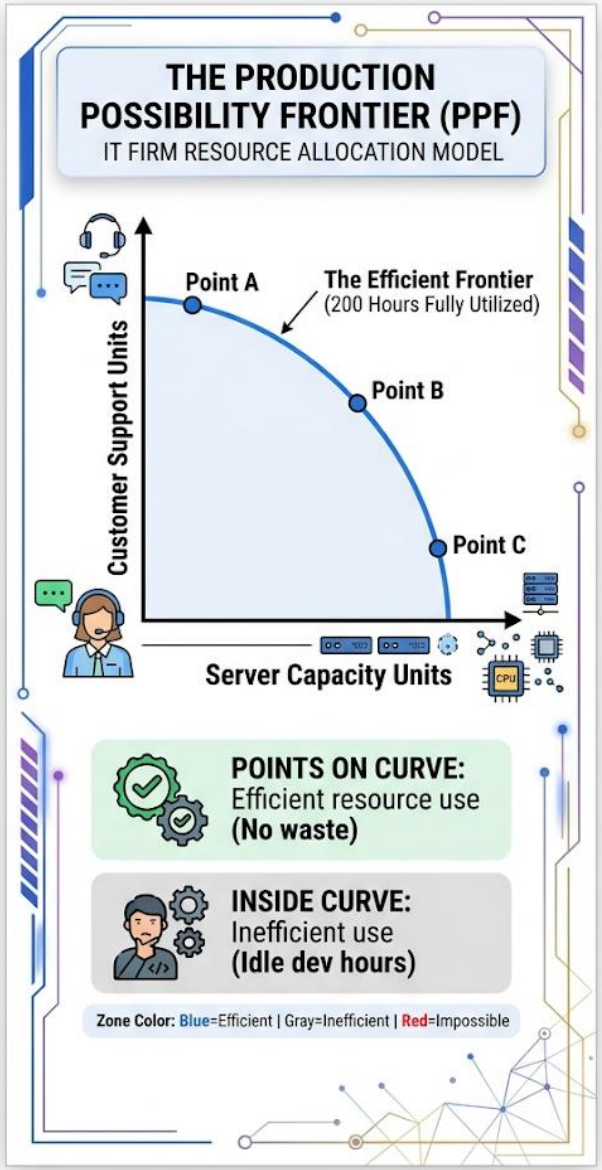
Modeling the Trade-off

- **The Zero-Sum Reality:** With a strict limit of 200 developer hours, every single hour assigned to Server Capacity is automatically subtracted from Customer Support.
- **The Linear Assumption:** If trade-offs were perfectly linear, transferring 10 hours of labor would always yield the exact same constant change in output.
- **Why Linear Models Fail:** In real-world software engineering, adding more developers to a single task does not scale output linearly due to communication overhead.
- **The Need for Advanced Modeling:** To accurately represent real economic and computational systems, we must use models that reflect varying productivity rates.



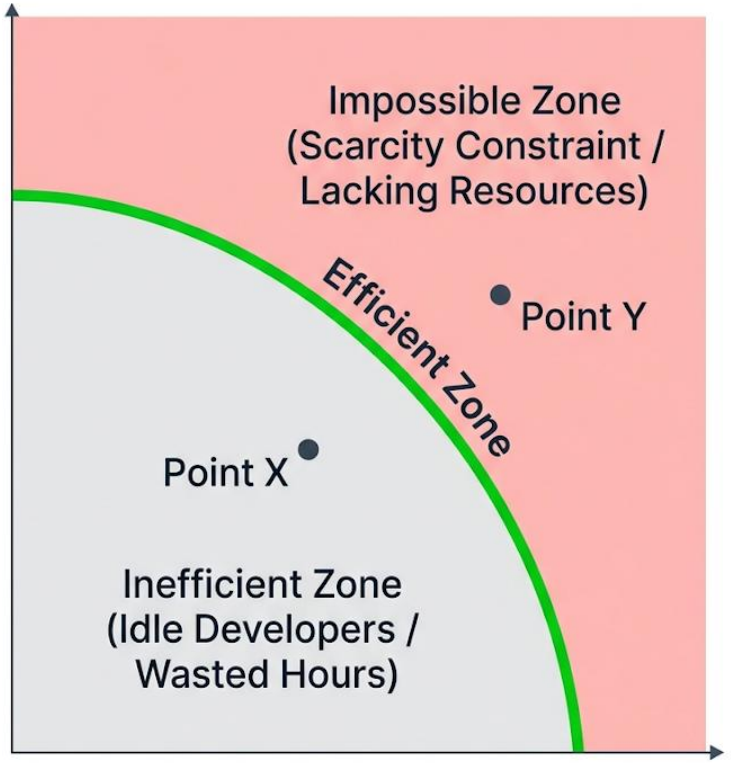
Introduction to the PPF

- **Defining the PPF:** The Production Possibility Frontier (PPF) is a standard economic model depicting all maximum output possibilities for two goods given fixed resources.
- **Visualizing Constraints:** The PPF curve maps the boundary between what is physically achievable by the firm and what is unattainable.
- **The Two Axes:** In our startup model, the X-axis represents units of Server Capacity produced, and the Y-axis represents units of Customer Support produced.
- **The Efficient Boundary:** Every point located directly on the curved PPF line represents a state where all 200 developer hours are fully and efficiently utilized.



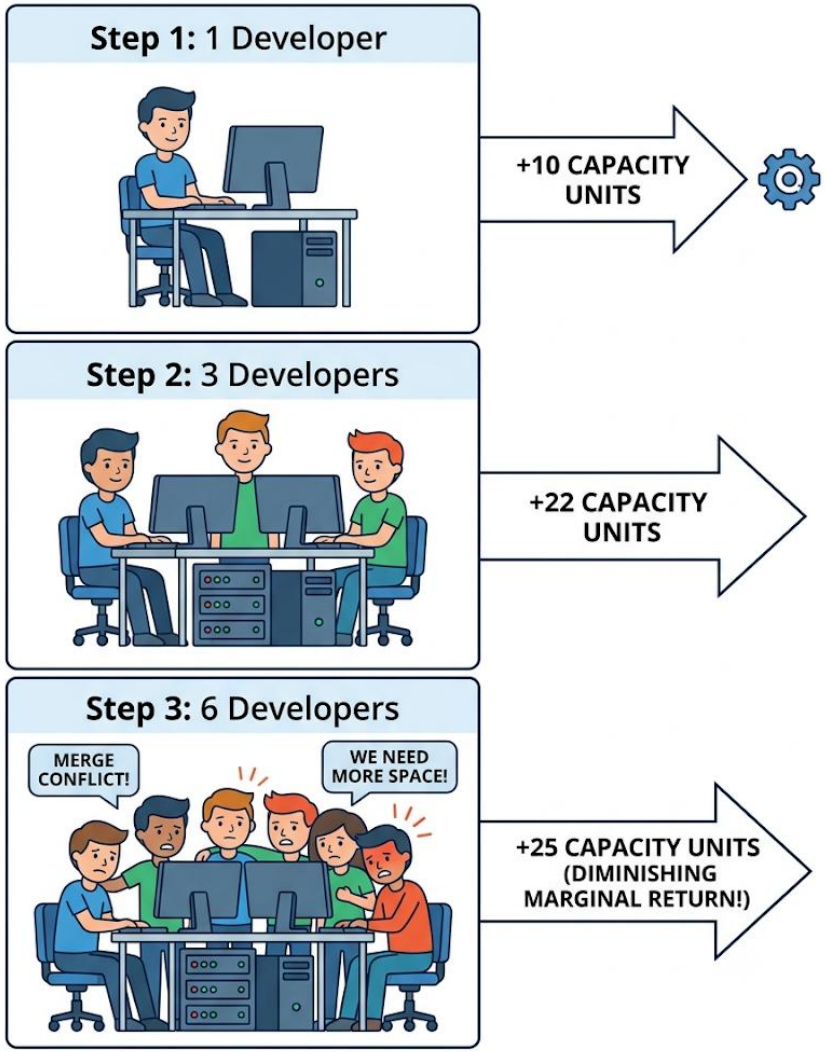
Mapping the Three Economic Zones

- **The Efficient Zone:** Points lying directly on the PPF curve indicate maximum operational efficiency, where no resources are wasted or left idle.
- **The Inefficient Zone:** Points located inside (below) the curve represent inefficiency, meaning developers are idle, poorly managed, or hours are being wasted.
- **The Impossible Zone:** Points located outside (above) the curve represent unattainable output levels that cannot be reached with the current 200-hour limit.
- **CTO Optimization Target:** As an optimizing manager, your primary operational goal is to ensure your startup always operates precisely on the efficient frontier.



The Law of Diminishing Returns

- **The Real-World Reality:** In both software development and economics, adding incremental units of a resource eventually yields diminishing marginal returns.
- **The Coding Analogy:** Assigning the 1st developer to a server module creates massive progress, but assigning a 10th developer creates communication bottlenecks and less output.
- **Why the PPF is Curved:** The PPF bows outward (concave to the origin) specifically because resources are not perfectly adaptable between different production tasks.
- **Marginal Trade-offs:** As you push to maximize Server Capacity, you must reassign developers who are better at Customer Support, increasing your opportunity costs.

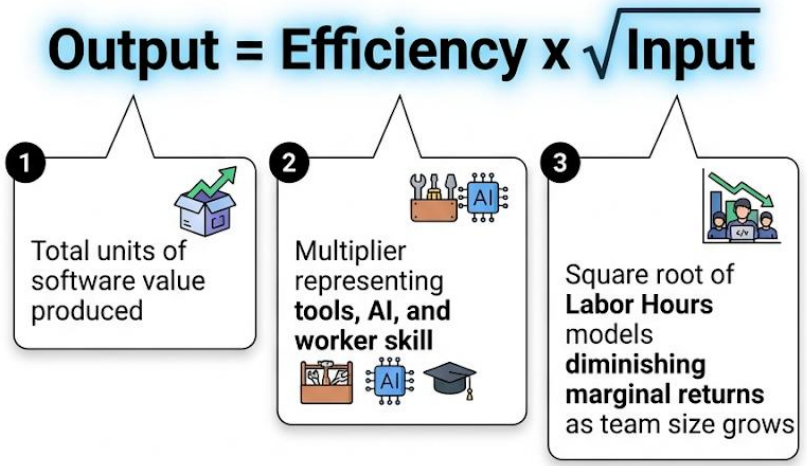


The Non-Linear Production Function

- **Mathematical Modeling:** To accurately capture diminishing returns in our Jupyter Notebook simulation, we model output using a non-linear square root function.
- **The Equation:** Output is calculated as:

$$Output = Efficiency \times \sqrt{\text{Input}}$$

- **The Input Variable:** The Input represents the raw number of developer labor hours allocated to that specific technical task.
- **The Efficiency Multiplier:** The Efficiency variable represents tool quality, worker skill, and management effectiveness, scaling the resulting output value.

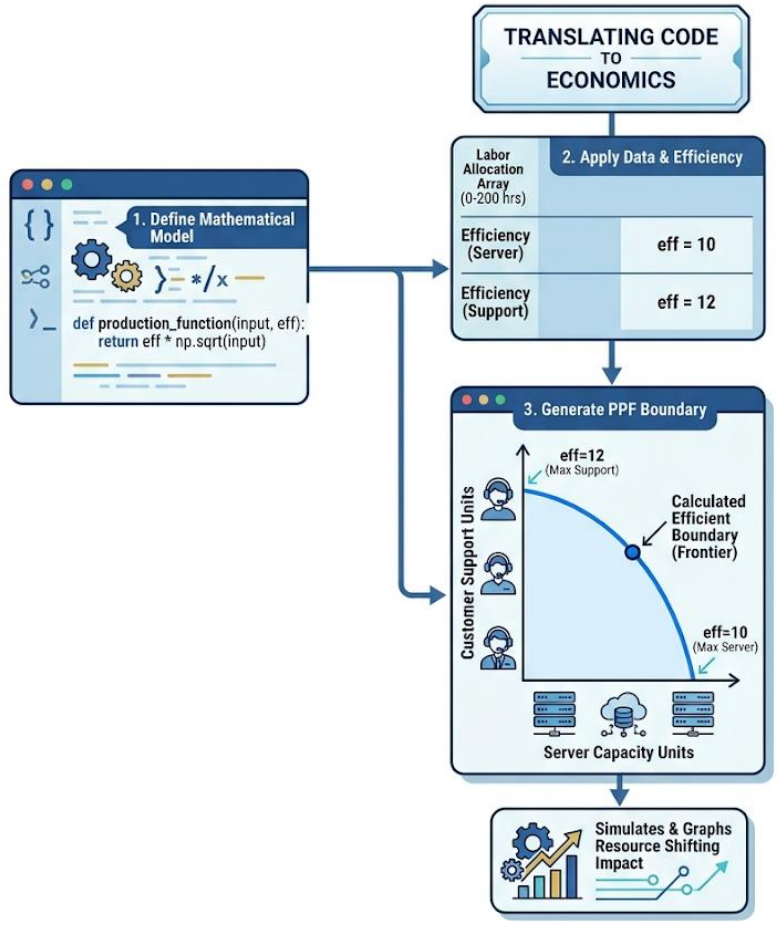


Translating Code to Economics

- **The Python Function:** In our tutorial, we implement the mathematical model using a simple Python function:

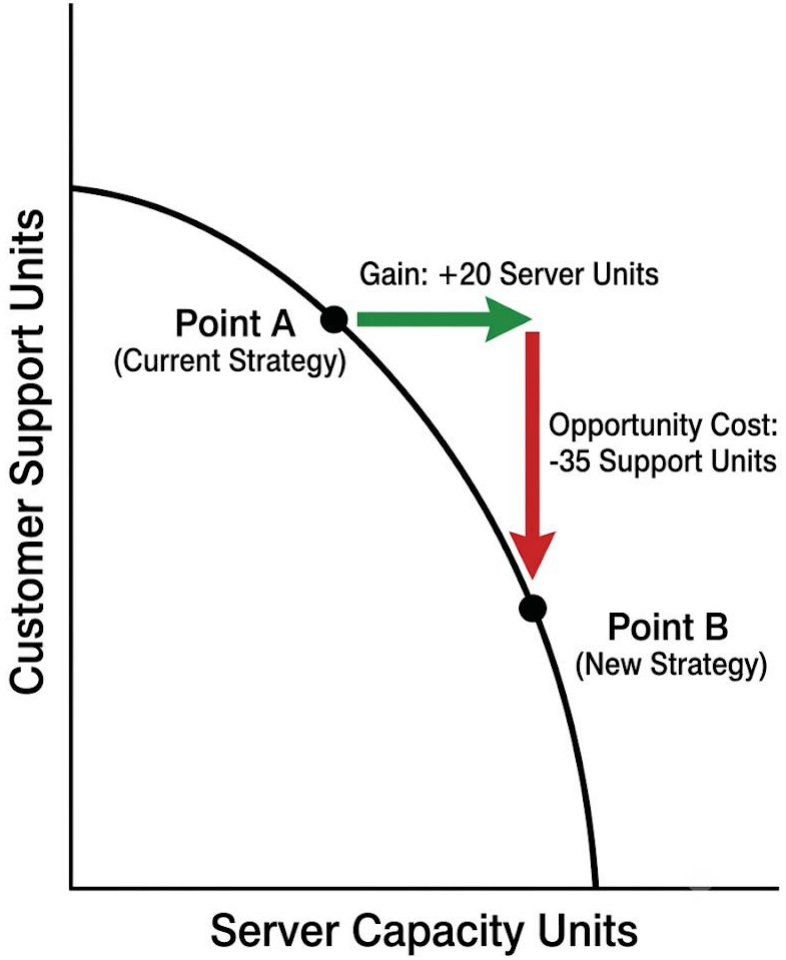
```
def production_function(resource_input, efficiency=10.0):
```

- **Task-Specific Efficiency:** We assign different baseline efficiency multipliers to reflect task difficulty: Server work uses `eff=10`, while Support work uses `eff=12`.
- **Generating Output Arrays:** By feeding an array of labor allocations from 0 to 200 hours into this function, we generate the exact coordinates of our PPF curve.
- **Algorithmic Precision:** This computational approach allows us to precisely simulate and graph how resource shifting impacts total firm productivity.



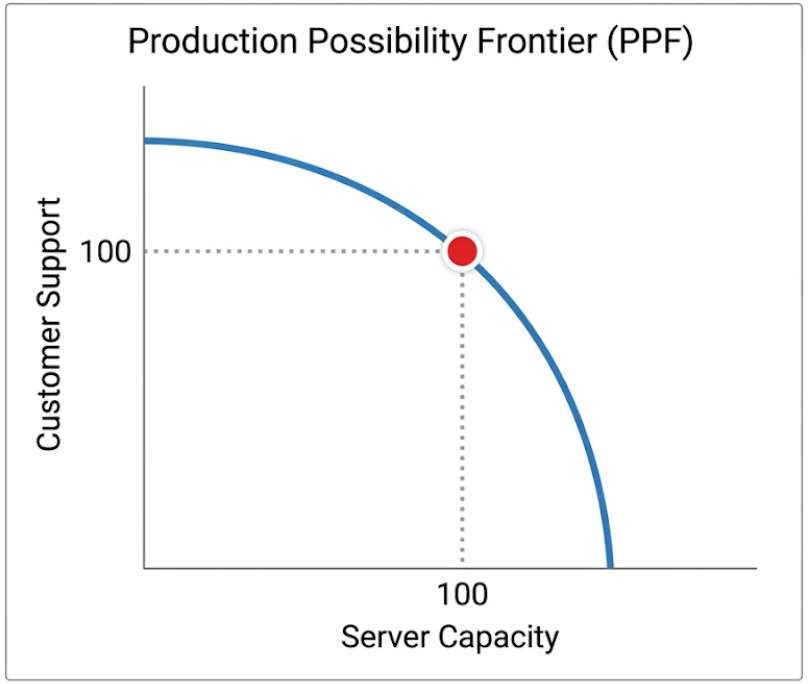
Visualizing Opportunity Cost on the PPF

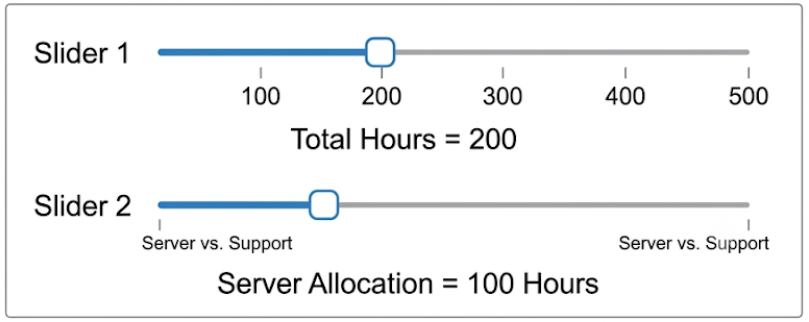
- **Moving Along the Curve:** When you change your startup's strategy by moving from Point A to Point B on the PPF, you reallocate developer hours.
- **Quantifying the Gain:** Moving rightward along the X-axis represents an intentional decision to increase production of Server Capacity units.
- **Quantifying the Cost:** The vertical drop down the Y-axis between the two points measures the exact loss in Customer Support capacity.
- **The Graphical Proof:** This vertical drop is the literal, visual quantification of your Opportunity Cost for choosing better server performance.



Interactive Resource Allocation Lab

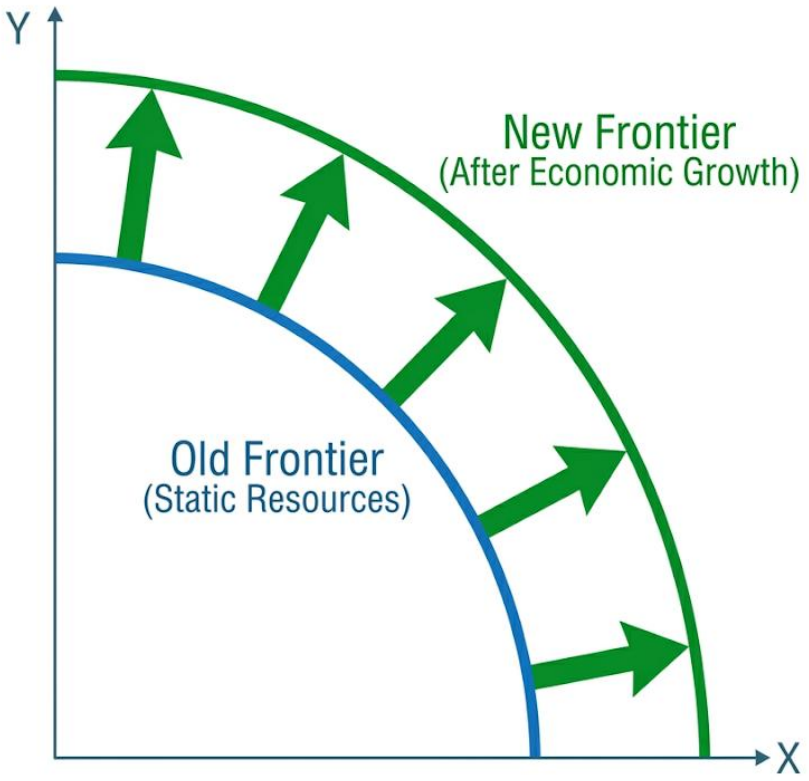
- **The Jupyter Notebook Tool:** Our tutorial features an interactive Python dashboard utilizing ipywidgets to let you dynamically act as TechNova's CTO.
- **Slider 1 - Scarcity Constraint:** The first interactive slider allows you to adjust total available developer labor hours between 100 and 500 hours.
- **Slider 2 - Strategic Choice:** The second slider controls your active allocation of those labor hours toward Server Capacity versus Customer Support.
- **Real-Time Feedback:** As you drag the allocation slider, a red marker moves along the PPF curve, calculating real-time output and opportunity costs.





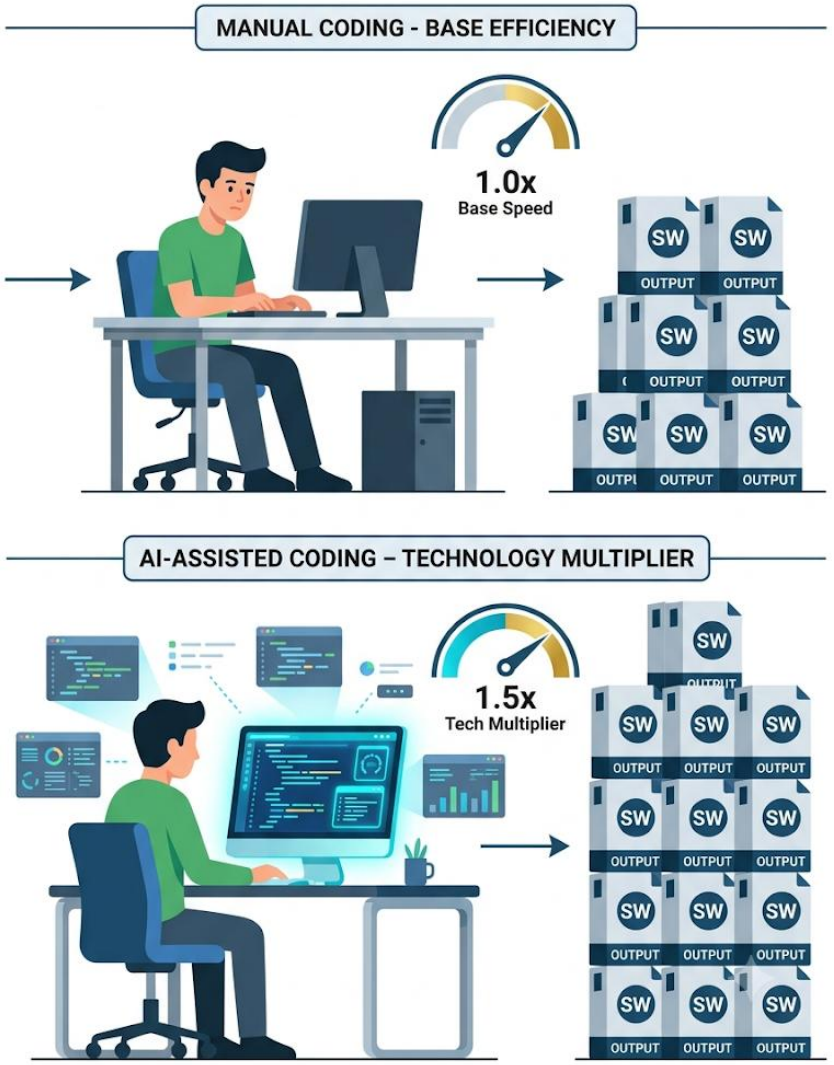
Breaking Constraints: Economic Growth

- **The Scarcity Barrier:** In a static system with fixed labor hours, it is physically impossible to produce more of both goods simultaneously.
- **The Growth Solution:** To break through the scarcity barrier and reach previously impossible output zones, an economy or firm must experience Economic Growth.
- **Two Paths to Growth:** Growth is achieved either by acquiring more raw resources (hiring more developers) or by increasing productivity (improving technology).
- **Shifting the Frontier:** When economic growth occurs, the entire PPF curve expands and shifts outward away from the origin.



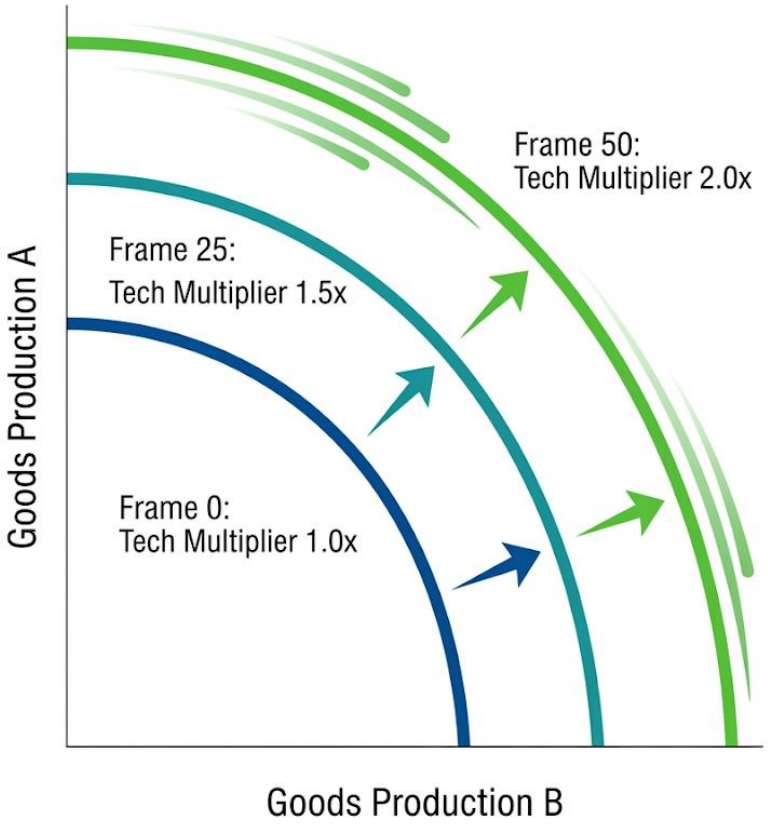
The Role of Technology in Economics

- **Technology as a Multiplier:** In economic theory, technological advancement acts as a direct multiplier on the efficiency of existing resources.
- **The Software Parallel:** Adopting modern AI coding assistants, automated testing suites, or better cloud tools makes developers significantly more productive.
- **Doing More With Less:** Better technology allows a startup to generate higher units of software output using the exact same 200 hours of human labor.
- **Algorithmic Impact:** In our production equation $Output = Efficiency \times \sqrt{\text{Input}}$, technology directly increases the Efficiency variable.



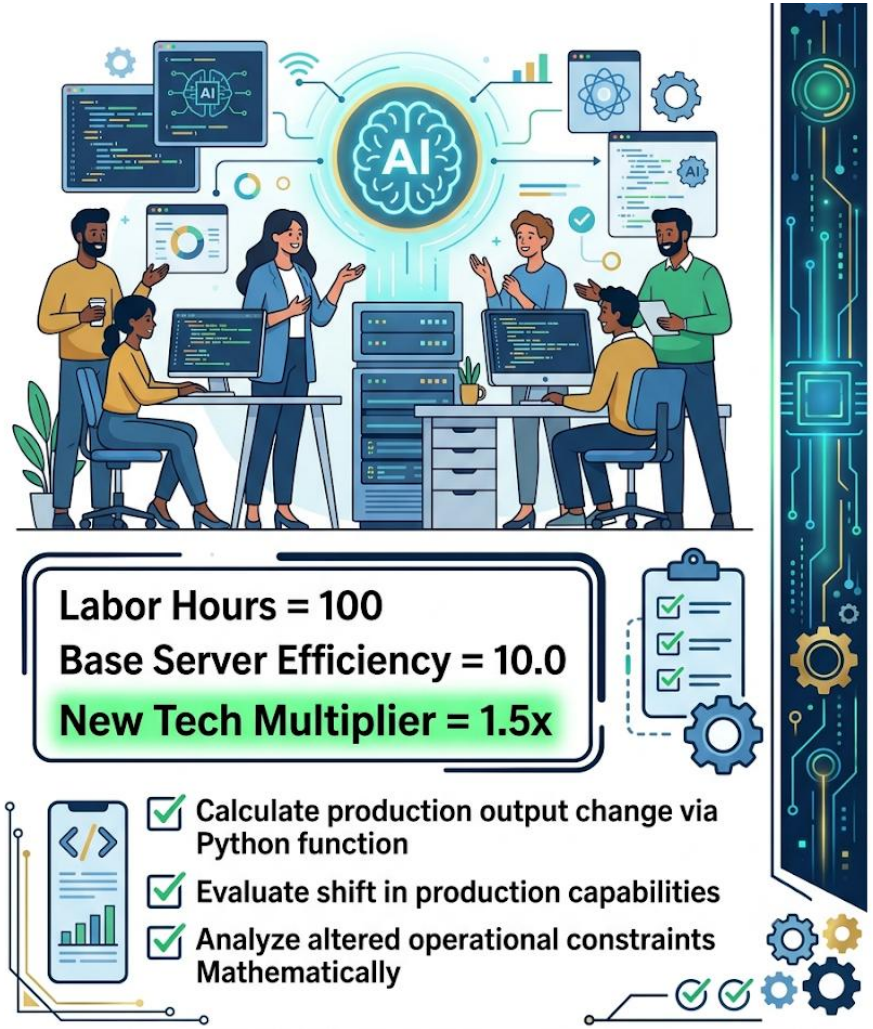
Simulating Growth with Python Animations

- **The Animation Script:** Our tutorial utilizes the matplotlib.animation library to dynamically simulate economic growth over 50 chronological frames.
- **The Evolving Multiplier:** In the code, the technology multiplier progressively increases from 1.0x (baseline) at Frame 0 to 2.0x at Frame 50.
- **Visualizing Expansion:** As the script executes, you observe the green PPF curve smoothly expanding outward into the previously impossible red zone.
- **Dynamic Decision Making:** This simulation proves that investing in technological tools fundamentally expands a firm's operational capabilities over time.



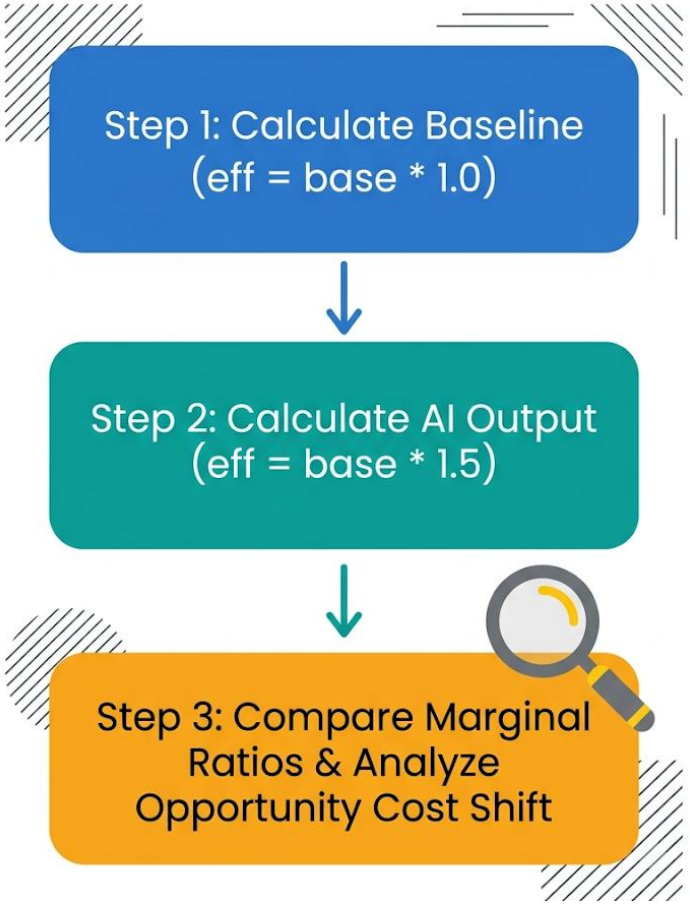
Homework Briefing: The AI Revolution

- **Assignment Context:** Your startup, TechNova, decides to adopt a cutting-edge AI coding assistant for all 5 software developers.
- **The Parameter Shift:** This technological adoption increases developer productivity by 50%, setting your new Technology Multiplier to 1.5.
- **Your Prerequisite Task:** You must use the provided Python production_function to analyze how this change impacts your production capabilities.
- **Analytical Focus:** The goal is not just running code, but mathematically evaluating how enhanced efficiency alters your operational constraints.



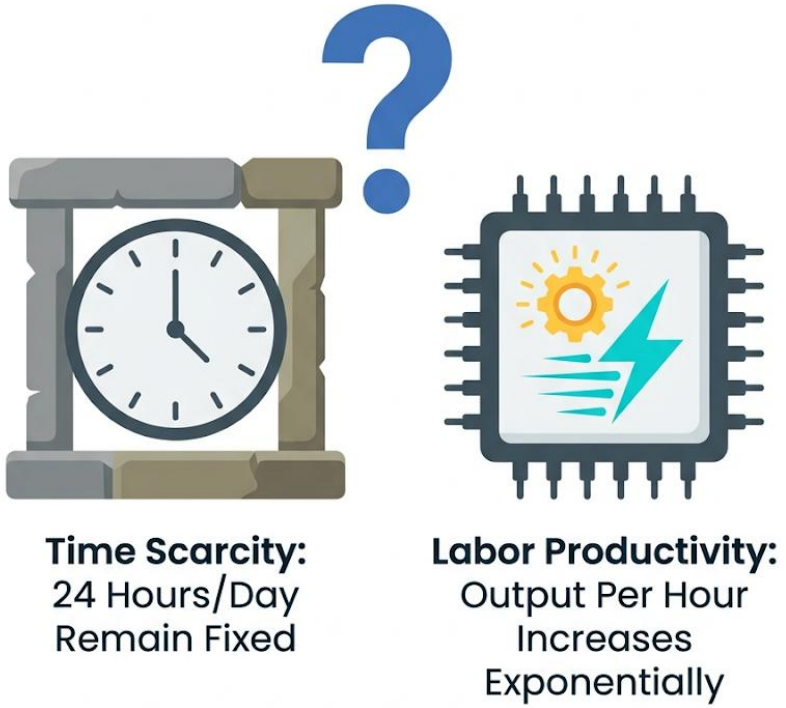
Homework Briefing: Calculating Opportunity Cost

- **The Analytical Question:** How does the new opportunity cost of producing one unit of Server Capacity compare to your original baseline?
- **Step 1 - Baseline Calculation:** You must first calculate total Server and Support outputs at the baseline efficiency multiplier of 1.0.
- **Step 2 - AI Calculation:** Next, re-calculate the exact outputs when the efficiency multiplier is increased to 1.5.
- **Step 3 - Comparative Analysis:** Compare the marginal trade-off ratios between the two scenarios to determine if the relative opportunity cost has shifted.



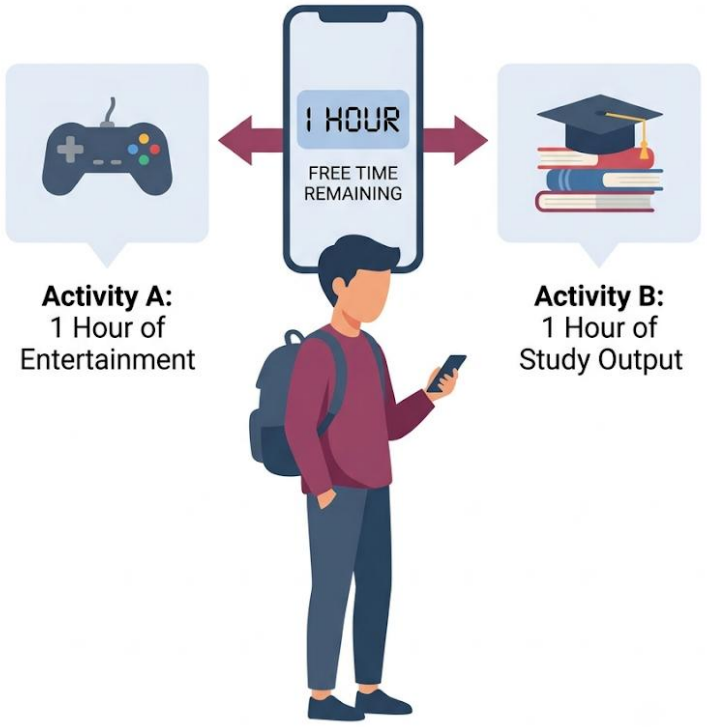
Homework Briefing: Critical Analysis

- **The Conceptual Prompt:** You must answer a deep theoretical question: Does adopting AI technology change the *Scarcity* of time, or the *Productivity* of time?
- **Evaluating Scarcity:** Consider whether giving your developers AI tools physically increases the number of hours in a day or days in a week.
- **Evaluating Productivity:** Consider how the AI tool affects the amount of software output generated per hour of human effort expended.
- **Writing Your Analysis:** Your submission should synthesize computational parameters with economic theory to provide a clear, logical justification.



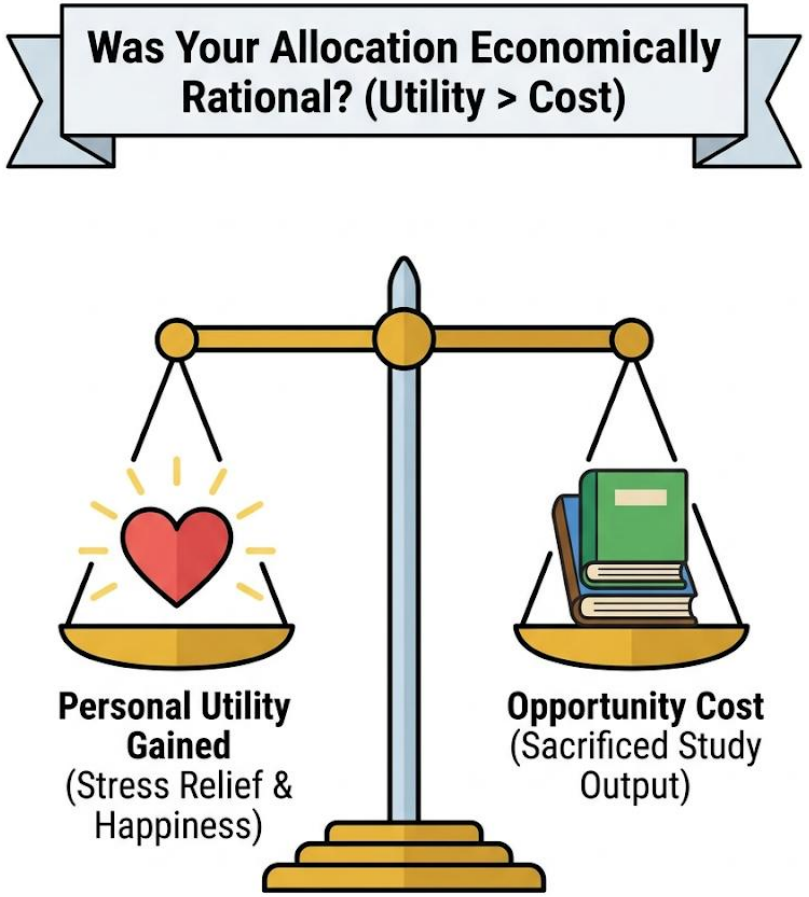
Homework Briefing: Daily Life Scarcity

- **The Real-World Application:** Economics is an operating system for daily life; to prove this, you must analyze a personal experience from this past week.
- **Identifying Scarcity:** Identify a specific instance where you faced a strict scarcity of time (e.g., having only 1 free hour before a deadline).
- **Defining the Choice:** Document the two competing activities you had to choose between, such as Activity A (Entertainment/Gaming) vs. Activity B (Studying/Work).
- **The Core Assignment:** You must formally calculate and describe the opportunity cost of the choice you made in terms of foregone output.



Homework Briefing: Rational Optimization

- **Evaluating Your Choice:** After identifying your personal opportunity cost, you must critically evaluate whether your decision was economically rational.
- **Applying Utility Theory:** Did choosing entertainment provide higher personal utility and stress relief than the study output you sacrificed?
- **The Goal of Reflection:** There is no "wrong" choice if the utility gained legitimately outweighed the value of the foregone alternative.
- **Course Synthesis:** This exercise trains you to continuously perform cost-benefit analyses on your most valuable scarce resource: your personal time.



Session Summary & Next Steps

- **Core Summary - Optimization:** Both computer science and economics focus on optimizing system performance under strict resource constraints.
- **Core Summary - Scarcity & PPF:** Scarcity forces trade-offs, and the Production Possibility Frontier visually maps the efficient boundaries of those choices.
- **Core Summary - Growth:** Technology acts as an efficiency multiplier, shifting the PPF curve outward and driving economic growth.
- **Action Items for Students:** Complete the Jupyter Notebook AI calculations, write your personal opportunity cost reflection, and prepare for Session 2!



Let’s check out our Notebook

